FINAL YEAR PROJECT - PART II

PROJECT ROADMAP AND GANTT CHART

Cost-Efficient Optimization of Open-Source LLM Agents for System-Level Tasks

PROJECT PERIOD
5 October - 27 November 2026

SUPERVISOR
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TEAM MEMBERS
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TARGET CHECKPOINT
Demonstrable 60% working prototype (planning target; not guaranteed)

Prepared for FYP 2 planning and progress tracking

1. Project Overview

This roadmap allocates eight weeks to three connected agent prototypes and a shared Ollama benchmarking workflow. Each member owns one module; the team shares model selection, evaluation, integration, and the final demonstration. The schedule is intentionally focused on testable core workflows rather than production-ready systems.

Project window	Team	Checkpoint
5 Oct - 27 Nov 2026	3 named members	60% working prototype target

System Workflow Diagram

INPUTS	>	THREE AGENTS	>	SHARED OLLAMA + BENCHMARK	>	DEMONSTRATION
Research papers Inventory and sales data 2D maps and obstacles		Research assistant Inventory agent Robot planning agent		Five model candidates Baseline vs optimized runs Quality, latency, token metrics		Three repeatable demos Acceptance evidence Limitations and next steps

The shared benchmark reports baseline and optimized results using the same task cases wherever possible. Each module also keeps its own task-quality checks.

2. Team Workstreams and Ownership

Member	Assigned workstream	Primary outputs
Khurram Rauf	Semantic Scholar AI Research Assistant and research-corpus quality pipeline	Verified paper list; Paper_ID metadata workbook; traceable PDF acquisition, naming, deduplication, integrity checks, ID checks, and reconciliation
Arzo Naveed	AI Inventory Management Agent for E-commerce	SQLite inventory and sales data; low-stock alerts; demand forecast; validated reorder recommendations; Streamlit demo
Momina Haroon	Robot Planning Agent - Autonomous 2D Robot Navigation Simulator	Grid simulator; start/goal/obstacle controls; A* routes; natural-language instructions; dynamic replanning demo
All members	Shared Ollama model study, benchmark pipeline, integration, and final review	Five-model shortlist; common task cases; baseline-versus-optimized results; integrated demo and report

Weekly Progress Review

Each owner updates planned versus completed tasks, demo evidence, benchmark coverage, blockers, and one measurable next outcome. Estimate progress from evidence rather than task count alone.

Week ending	Owner / workstream	Evidence or demo	Blocker / next outcome
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

3. Weekly Roadmap and Milestones

Week / dates	Khurram Rauf - Research Assistant	Arzo Naveed - Inventory Agent	Momina Haroon - Robot Agent	Shared milestone / deliverable
W1 | 5-9 Oct	Verify the starting reference list; define verification checklist and Paper_ID rules. Draft metadata columns.	Define products, stock, sales fields and SQLite schema; prepare sample cases.	Define grid, coordinates, obstacle rules, and simulator acceptance cases.	Kickoff; scope and acceptance checks frozen; local hardware/Ollama check started.
W2 | 12-16 Oct	Build Python metadata extraction sheet with Paper_ID, Title, Bibtex_Citations, SS_Link, DOI, and Paper-Downloaded-Status.	Create SQLite schema and seed product/stock/sales data; specify validated inventory actions.	Build 2D grid simulator with start, destination, and obstacle selection.	Select five suitable Ollama models for local hardware; define shared input/output contract and benchmark cases.
W3 | 19-23 Oct	Test extraction; use Python/Ollama to suggest filename-to-Paper_ID matches; human-check ambiguous matches, rename files, and update status with an audit map. Begin acquisition.	Implement inventory data workflow and low-stock alerts; add guardrails to inventory actions.	Implement A* path calculation and route-validity checks.	First end-to-end baseline path in each module; smoke-test common Ollama interface.
W4 | 26-30 Oct	Use Python clients to find remaining papers; use Ollama for metadata normalization and matching suggestions, then verify each record and access route. Begin DOI/title duplicate detection.	Load sales history; build a simple demand forecast baseline and validation split.	Add Ollama-based natural-language navigation instruction parsing; test fixed maps.	Common benchmark harness captures latency, token usage, task success, and module quality.
W5 | 2-6 Nov	Refining 1: remove duplicate papers and merge IDs to a canonical Paper_ID. Refining 2: flag approx. 1 KB/invalid PDFs, validate, and remove confirmed corrupt files and their metadata rows.	Evaluate forecast error; calculate reorder quantities; test low-stock and action accuracy.	Add dynamic obstacles and trigger replanning; validate collision-free paths.	Run baseline experiments across the five-model shortlist where hardware permits; log actual coverage.
W6 | 9-13 Nov	Refining 3: find one Paper_ID assigned to distinct papers. Refining 4: compare PDF count with metadata rows and reconcile filenames to Paper_ID.	Complete Streamlit inventory workflow and edge-case checks.	Complete simulator workflow and path metrics; add no-path and obstacle tests.	Finish repeatable benchmark runs; consolidate baseline results and integration defects.
W7 | 16-20 Nov	Verify repaired corpus and produce exception/reconciliation report.	Tune prompts/context; reduce unnecessary model calls while preserving validated stock logic.	Tune instruction prompt/context; reduce unnecessary LLM calls; stress-test replanning.	Matched baseline-versus-optimized experiments; integrate the three demo flows; freeze feature scope.
W8 | 23-27 Nov	Demonstrate clean metadata/PDF reconciliation on the fixed test corpus; prepare module evidence.	Demonstrate low-stock alert, forecast, and validated reorder suggestion; prepare evidence.	Demonstrate natural-language task to A* route and dynamic obstacle replanning; prepare evidence.	Acceptance review, final demo, report/slides, run instructions, benchmark summary, limitations, and next-phase backlog.

4. Research Corpus Workflow - Khurram Rauf

Stage	Planned procedure	Deliverable / check
Verification	Verify the starting article list before downstream processing; check title/DOI and intended inclusion. Mark uncertain records for review.	Verified article queue with a clear decision/status per record.
Metadata extraction	Use a Python-based tool to construct Meta_Data.xlsx with Paper_ID, Title, Bibtex_Citations, SS_Link, DOI, and Paper-Downloaded-Status.	Consistent metadata workbook and reproducible extraction log.
Existing file-ID renaming	Use a Python/Ollama-assisted title or DOI match to suggest filename-to-Paper_ID links; human-review ambiguous matches. Rename existing PDFs, update Paper-Downloaded-Status, and retain a rename/audit map. Use a dry run before batch changes.	No ambiguous overwrite; each rename can be traced to its original filename and match evidence.
Acquisition	Use Python clients to search Semantic Scholar, OpenAlex, open-access repositories, publisher pages, author manuscripts, or institutionally authorized library access. Ollama may normalize metadata or suggest record matches; verify the paper and access route. Save with Paper_ID and update status.	Acquisition log distinguishes downloaded, unavailable, and needs-review records.
Refining 1 - duplicates	Detect repeated DOI/title even under different Paper_IDs; choose a canonical Paper_ID, merge aliases/links, and remove duplicate files.	Duplicate report and canonical-ID mapping.
Refining 2 - corrupt PDFs	Flag files around 1 KB or failing PDF readability checks; confirm corruption, remove the confirmed bad file and its matching metadata row, and record the action.	Corruption report and synchronized file/metadata cleanup.
Refining 3 - ID collision	Find a Paper_ID associated with different titles or DOIs; resolve the ID mapping before further use.	Unique Paper_ID-to-paper mapping.
Refining 4 - reconciliation	Compare PDF count with metadata rows; parse filenames and match Paper_IDs; list missing PDFs, orphan PDFs, and metadata-only rows for resolution.	Final reconciliation report with counts and exception list.

5. Gantt Chart - 5 October to 27 November 2026

Task bars show planned work weeks; dependencies appear as task IDs. Blue = Khurram Rauf, green = Arzo Naveed, orange = Momina Haroon, and teal = shared work. The editable chart follows a GanttPRO-style work breakdown with named owners and visible predecessors.

ID	Work package	Owner	Depends on	W1 5-9 Oct	W2 12-16 Oct	W3 19-23 Oct	W4 26-30 Oct	W5 2-6 Nov	W6 9-13 Nov	W7 16-20 Nov	W8 23-27 Nov
T01	Kickoff, scope, acceptance tests	All	-								
T02	Verify article list and define Paper_ID rules	Khurram	T01								
T03	Select five Ollama models and define benchmark	All	T01								
T04	Metadata workbook and Python extraction	Khurram	T02								
T05	Rename existing PDFs and update download status	Khurram	T04								
T06	Acquire remaining aut horized/open-access PDFs	Khurram	T02-T04								
T07	Refinements 1-4 and corpus reconciliation	Khurram	T06								
T08	Inventory schema, SQLite, and sample data	Arzo	T01								
T09	Low-stock alerts and validated actions	Arzo	T08								
T10	Sales data and demand forecast evaluation	Arzo	T08								
T11	Reorder workflow, quantities, and Streamlit UI	Arzo	T09,T10								
T12	2D grid simulator and scenario controls	Momina	T01								
T13	A* pathfinding and route validation	Momina	T12								
T14	Natural-language tasks and dynamic replanning	Momina	T03,T13								
T15	Common benchmark harness and fixed task cases	All	T03								
T16	Baseline experiments and metric capture	All	T06,T09,T13,T14,T15								
T17	Prompt/context optimization and fewer LLM calls	All	T16								
T18	Integration, acceptance checks, final report/demo freeze	All	T07,T11,T14,T17								

6. Shared Ollama Benchmark and Evaluation

Select five suitable Ollama models during Week 2 based on available RAM/VRAM, model size/quantization, local inference speed, task fit, and licensing/access constraints. Record exact model tags and runtime settings. Use a small smoke set across all five first; run repeated full comparisons on the models the hardware can support, and disclose the actual coverage.

Metric	Common measurement	Research assistant	Inventory agent	Robot agent
Task success	Pass rate on fixed acceptance cases	Correct paper record and clean ID/file reconciliation	Correct alert/reorder action	Goal reached; valid/no-path result
Quality / accuracy	Task-specific rubric, scored with evidence	Title/DOI/link/citation and metadata completeness	Inventory-action accuracy; demand-forecasting MAE or MAPE	Path validity, path length, collision count
Latency	End-to-end wall-clock time; report median and p95 when repeats allow	Time per query / corpus operation	Time per forecast/reorder workflow	Planning and replanning latency
Token usage	Prompt plus generated tokens, or consistent runtime proxy	Tokens per completed research task	Tokens per inventory action	Tokens per navigation task
Cost efficiency	Compare success/quality alongside latency and token usage	No quality loss in metadata steps	Forecast/action quality retained	Valid path and safety maintained

Baseline-versus-Optimized Experiment

- Freeze representative test cases before tuning; reuse the same cases, machine, model tag, and settings for each paired comparison.
- Baseline: current prompt/context and call pattern. Optimized: shorter relevant context, clearer prompt structure, cached/reused results, and deterministic checks where an LLM call is unnecessary.
- Track task success and task-specific quality together with latency and token use. A faster run that reduces correctness is not a successful optimization.
- Keep a run log with date, model tag, hardware, settings, case ID, outputs, measured values, and errors.

7. What “60% Working” Means by 27 November

Treat 60% as a demonstrable scope checkpoint, not a promise of completion or a production-readiness claim. Assess it against a fixed acceptance list and show the evidence:

- Research assistant: verify the article set, create the required metadata workbook, demonstrate Paper_ID naming/download-status updates, and run all four refinement checks on a representative fixed corpus with an exception report.
- Inventory agent: load sample products, stock, and sales into SQLite; show a low-stock alert, a basic demand forecast with reported error, and a validated reorder recommendation in the demo.
- Robot agent: show the 2D grid, start/goal/obstacles, A* route, natural-language instruction handling, and at least one dynamic-obstacle replanning case.
- Shared evaluation: select and document five Ollama models; run baseline and optimized comparisons for available model/task combinations; report actual coverage and all four shared metrics.
- Overall: at least 60% of the pre-agreed core acceptance cases pass, with each module represented; reproducible run instructions, test evidence, known limitations, and next-phase work are documented.

This target is tight for eight weeks and depends on local hardware, document/source availability, data quality, and timely integration. Report the measured completion percentage honestly if any acceptance criteria remain blocked.

8. Dependencies, Risks, and Controls

Risk / dependency	Control and owner
Model downloads or local inference exceed hardware capacity	Khurram leads the Week 1 hardware check; select five suitable candidates in Week 2; use smoke tests first and disclose which runs were feasible.
Paper access is unavailable or metadata is incomplete	Khurram records unavailable items and tries Semantic Scholar, OpenAlex, open-access repositories, publisher pages, author manuscripts, and institutionally authorized access.
Batch renaming or deduplication creates data loss	Khurram uses a dry run, a reversible rename map, canonical-ID mapping, and an exception log before cleanup.
Inventory forecast has too little history	Arzo uses a clearly labeled sample dataset, a simple baseline forecast, and a fixed validation split; state the data limitation.
Robot LLM output is unsafe or invalid	Momina validates instructions against simulator constraints; A* and path checks remain authoritative for route validity and collisions.
Integration slips into final week	Freeze interfaces in Week 2; run module smoke integration in Week 3; hold weekly demos; freeze features by 20 Nov.